previously. In particular, the Boost Thread Library has been used as the primary model on which the new library is based, with many of the classes sharing their names and structure with the corresponding ones from Boost. As the new standard has evolved, this has been a two-way flow, and the Boost Thread Library has itself changed to match the C++ Standard in many respects, so users transitioning from Boost should find themselves very much at home.

Concurrency support is just one of the changes with the new C++ Standard —as mentioned at the beginning of this chapter, there are many enhancements to the language itself to make programmers' lives easier. Although these are generally outside the scope of this book, some of those changes have had a direct impact on the Thread Library itself and the ways in which it can be used. Appendix A provides a brief introduction to these language features.

The support for atomic operations directly in C++ enables programmers to write efficient code with defined semantics without the need for platform-specific assembly language. This is a real boon for those trying to write efficient, portable code; not only does the compiler take care of the platform specifics, but the optimizer can be written to take into account the semantics of the operations, thus enabling better optimization of the program as a whole.

1.3.3. Efficiency in the C++ Thread Library

One of the concerns that developers involved in high-performance computing often raise regarding C++ in general, and C++ classes that wrap low-level facilities—such as those in the new Standard C++ Thread Library specifically is that of efficiency. If you're after the utmost in performance, then it's important to understand the implementation costs associated with using any high-level facilities, compared to using the underlying low-level facilities directly. This cost is the *abstraction penalty*.

The C++ Standards Committee has been very aware of this when designing the C++ Standard Library in general and the Standard C++ Thread Library